Note

A non-zero matrix is in a row-echelon form if all zero rows occur as bottom rows of the matrix, and if the first non-zero element in any lower row occurs to the right of the first non-zero entry in the higher row.

The following matrices are in row-echelon form:(i) $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$,(ii) $\begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 0 & 2 & 8 \\ 0 & 0 & 0 & 6 \end{bmatrix}$

Consider the matrix in (i). Go up row by row from the last row of the matrix. The third row is a zero row. The first non-zero entry in the second row occurs in the third column and it lies to the right of the first non-zero entry in the first row which occurs in the second column. So the matrix is in row-echelon form.

Consider the matrix in (ii). Go up row by row from the last row of the matrix. All the rows are non-zero rows. The first non-zero entry in the third row occurs in the fourth column and it occurs to the right of the first non-zero entry in the second row which occurs in the third column. The first non-zero entry in the second row occurs in the third column and it occurs to the right of the first non-zero entry in the first row which occurs in the first column. So the matrix is in row-echelon form.

The following matrices are not in row-echelon form:

(i) $\begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 5 \\ 0 & 1 & 0 \end{bmatrix}$, (ii) $\begin{bmatrix} 0 & 3 & -2 \\ 5 & 0 & 0 \\ 3 & 2 & 0 \end{bmatrix}$.

Consider the matrix in (i). In this matrix, the first non-zero entry in the third row occurs in the second column and it is on the left of the first non-zero entry in the second row which occurs in the third column. So the matrix is not in row-echelon form.

Consider the matrix in (ii). In this matrix, the first non-zero entry in the second row occurs in the first column and it is on the left of the first non-zero entry in the first row which occurs in the second column. So the matrix is not in row-echelon form.

Method to reduce a matrix $\left[a_{ij}\right]_{m \times n}$ to a row-echelon form.

Step 1

Inspect the first row. If the first row is a zero row, then the row is interchanged with a non-zero row below the first row. If a_{11} is not equal to 0, then go to step 2. Otherwise, interchange the first row R_1 with any other row below the first row which has a non-zero element in the first column; if no row below the first row has non-zero entry in the first column, then consider a_{12}. If a_{12} is not equal to 0, then go to step 2. Otherwise, interchange the first row R_1 with any other row below the first row which has a non-zero element in the second column; if no row below the first row has non-zero entry in the second column, then consider a_{13}. Proceed in the same way till we get a non-zero entry in the first row. This is called **pivoting** and the first non-zero element in the first row is called the **pivot** of the first row.

Step 2

Use the first row and elementary row operations to transform all elements under the pivot to become zeros.

Step 3

Consider the next row as first row and perform steps 1 and 2 with the rows below this row only. Repeat the step until all rows are exhausted.

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